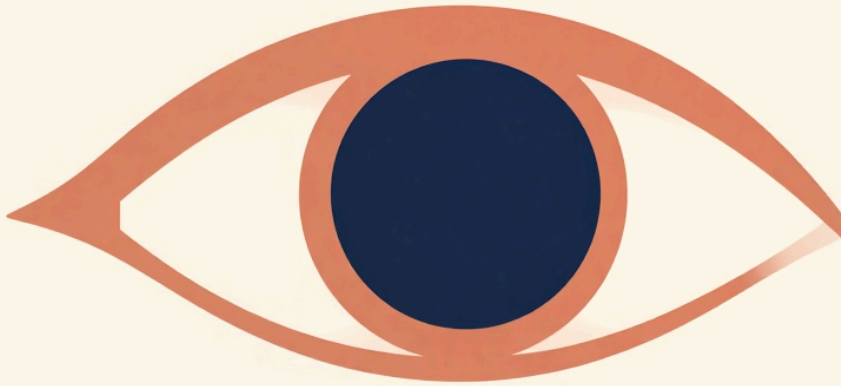




PAPER EXPLAINED

Detect blinks faster with linear attention transformers

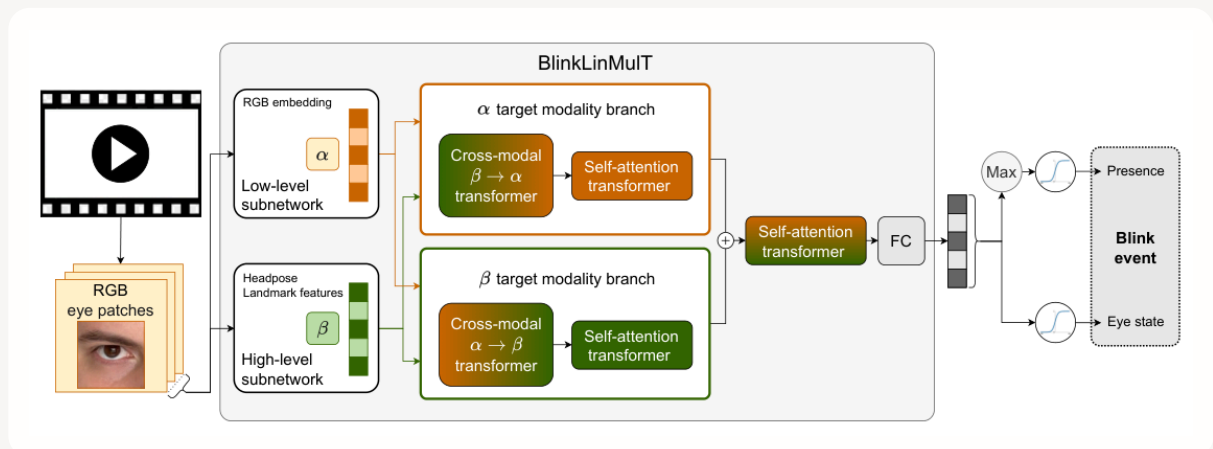
BlinkLinMulT: Transformer-Based Eye Blink Detection

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https://adamfodor.com/pdf/2023_Fodor_Adam_MDPI_BlinkLinMulT.pdf

Eye Blink Detection for Safety and Security

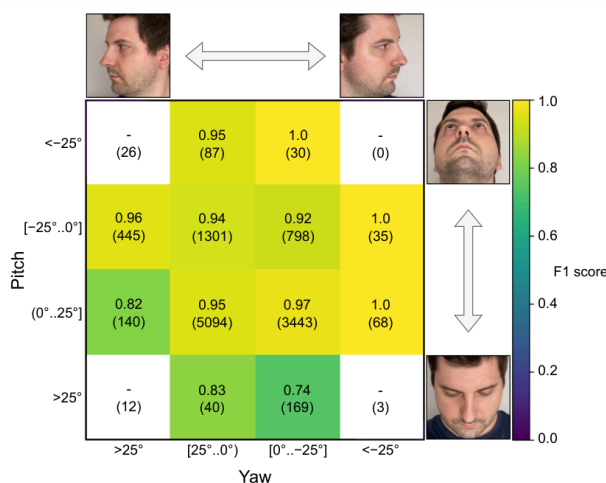
- Detecting eye blinks solves facial analysis tasks like deep fake detection and driver drowsiness monitoring.
- Methods are categorized into feature-based, appearance-based, and motion-based approaches.
- The approach is evaluated on blink presence detection and eye state recognition tasks using multiple public benchmark datasets.



BlinkLinMulT: a multimodal transformer architecture for blink presence detection and frame-wise eye state recognition.

Challenges in Handling Noisy Data and Short Blinks

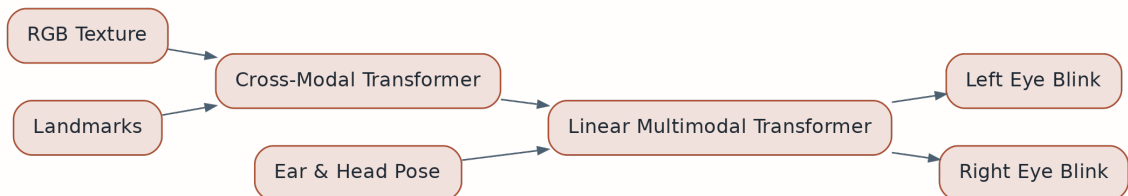
- Existing face recognition and frame-wise classification methods struggle with short blinks or noisy data.
- The CEW dataset lacks pre-defined splits, requiring a 10-fold cross-validation scheme to measure performance.
- Extreme head pose angles cause most errors due to eyelid visibility and landmark precision issues.



Head pose angle dependence of BlinkLinMulT in the case of blink presence detection task.

BlinkLinMulT: Linear Attention for Multi-Modal Input

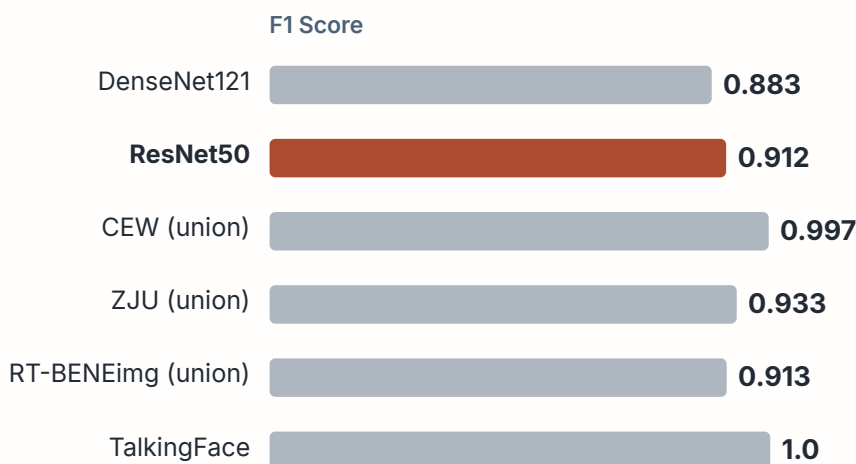
- Modified multi-modal transformer considers RGB texture, iris and eye landmarks, ear, and head pose angles.
- Cross-modal transformers translate between landmark features and RGB texture embeddings.
- Linear multimodal transformer estimates blinks for each eye independently.



BlinkLinMulT architecture: cross-modal transformers fuse RGB and landmark features, which feed into a linear multimodal transformer estimating blinks per eye.

Superior F1 Scores on RT-BENEimg and TalkingFace

- ResNet50 achieved a 0.912 F1 score on RT-BENEimg, outperforming DenseNet121's 0.883.
- Training on the union of datasets improved F1 scores: CEW rose to 0.997, ZJU to 0.933, and RT-BENEimg to 0.913.
- A model achieved a perfect 1.0 F1 score on the TalkingFace dataset for blink presence detection.



Superior F1 scores on RT-BENEimg and TalkingFace datasets